

Home with the Family

Mark Jordan - 10/16/25

Home with the family

We create a highly believable, private VR living-room where invited family can meet as tracked avatars to watch streaming content in a windowless browser, play tabletop games on the kitchen counter, and share presence across distances using light-weight WebXR clients and Gaussian splat rendering for realistic soft geometry.

Synopsis

Short narrative of the experience: user enters their home-world via headset or WebXR-capable browser, authentic avatar mapping preserves body and hand gestures, dynamic windowed media replaces the TV for shared streaming, and the counter can spawn interactive tabletop apps (cards, chess). Networked state, voice, and spatial audio make conversations feel natural.

Inspired by Gordon Ramsay's kitchen tour



<https://medium.com/@zoltanfood/i-was-walking-into-gordon-ramsays-kitchen-1cd979abac92>

Inspired by Little Tokyo model



What is authentic about my XR project

Real home geometry and layout derived from your bedroom footage
→ creates emotional authenticity.

Private invite model (family-only).

Real gestures and small social actions (handing a card, leaning across the counter).

Familiar activities (watching a show, playing chess) rather than abstract interactivity.

XR interaction

Enter / exit the home (door UI + invite link)

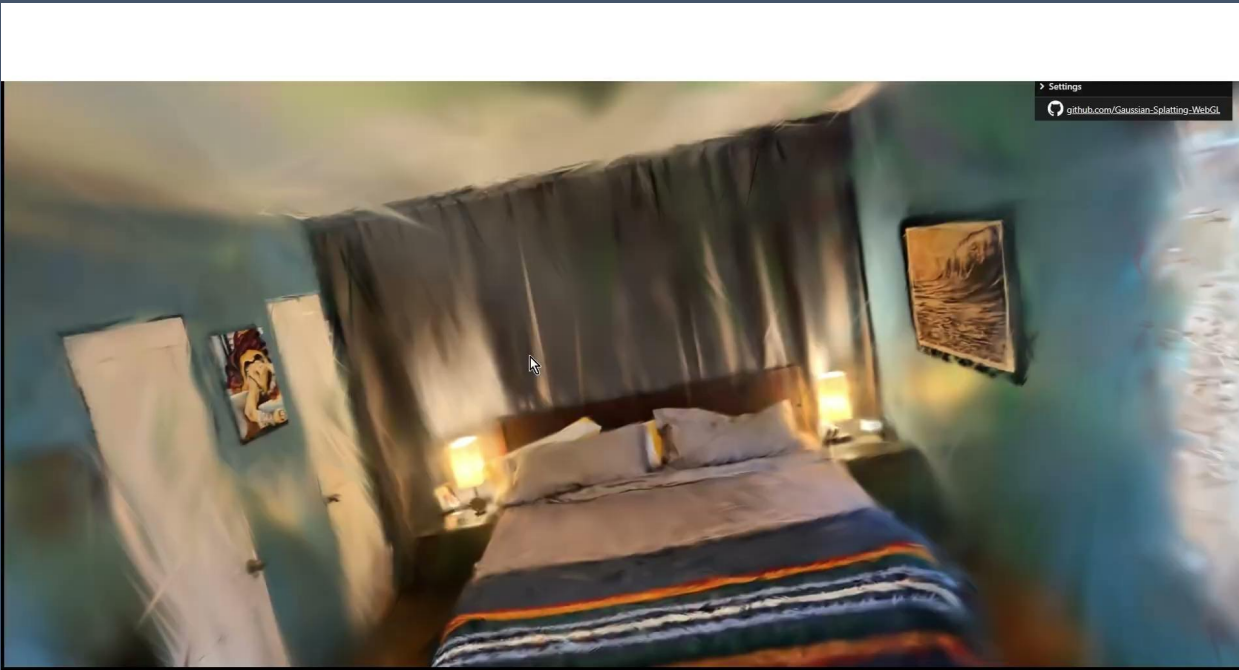
Screen-sharing: windowless Chrome-like in-world browser for streaming (host invites)

Tabletop apps: spawn chess/cards on counter, grab & place objects

Hand gestures: pointing, thumbs-up, clapping, passing items (via physics / snap-to)

Voice & spatial audio, lip-sync approximation on avatars

Genre of the project



Tone of the project

Warm and inviting atmosphere that feels like home.

Bright, happy mood that encourages connection and laughter.

Relaxed pacing with friendly, natural interactions.

Focus on comfort, positivity, and togetherness rather than technology.

Sound design

Embedded video audio plays naturally from the chromeless browser window.

Continuous ambient music fills quiet spaces so silence is not defining.

Subtle room reverb matched to the living room's size for realism.

Spatialized voices with near-field occlusion (voices soften when someone moves behind furniture or whispers close).

Interaction sounds—card shuffle, cup clink, and soft UI chimes—create natural feedback and a lived-in feel.

Dialogue/voice over?

Minimal: short onboarding VO for first-time guests (“Welcome — you are joining Mark’s living room”), otherwise rely on natural speech and UI hints.

Storyboards – Concept Design – Levels



Characters?

Avatars:
stylized-but-accurate:
full-body with
expressive hands.
Minimal facial detail
with lip-sync and
eye-look.



Environments

Primary environment:
photoreal-ish living room
generated via Gaussian
splats using your bedroom
footage as the source.
Fallback stylized mesh
environment for low-end
clients.



Technical architecture

Client: WebXR-capable browser + native headset apps (fallbacks for 2D WebRTC viewers).

Rendering: Gaussian splats (point-splat renderer) for environment + baked light probes for local lighting.

Networking: WebRTC for low-latency audio + WebSocket / WebRTC data channels for synchronized object state (games, browser controls).

Avatar tracking: WebXR hand tracking / headset tracking; body mapping via inverse kinematics on headset + two controllers (or full-body optical tracking if available).

Media: Windowless browser (in-world Chromium iframe) uses host's stream + synced play/pause via data channel.

Thank You!

Mark Jordan
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